

Web Technologies Project Report: Coffi Coffee Shop (Alikhan Bokeikhan Branch)

Course: Web Technologies / Frontend Fundamentals

Repository URL: <https://github.com/jasulansam/web>

Team Members:

- Samrat Zhasulan
- Zhaksylyk Adilet

1. Project Overview

- **Objective:** Development of a 6-page accessible website for the Coffi coffeehouse (Alikhan Bokeikhan St 10, Astana) adhering strictly to semantic HTML5 standards without external styling or client-side scripting frameworks.
- **Core Architectural Rules:**
 - Strict landmark hierarchy (<header>, <nav>, <main>, <footer>).
 - Pure semantic structure (no inline styling, zero presentation attributes like border="1" or align).
 - Tabular data formatting using accessible table structures (<table>, <caption>, <thead>, <tbody>, and th scope).
 - Form inputs configured with explicit label associations, logical field grouping (<fieldset>, <legend>), and native HTML5 validation constraints across two functional forms.

2. Task A — Anatomy of Coffi Web Architecture & Semantic Planning

2.1 Head Elements and Metadata Inspection (Coffi Coffee Shop)

- **DOCTYPE:** <!DOCTYPE html> (HTML5 doctype ensuring modern standard rendering mode)
- **Language:** lang="en" in <html lang="en"> (specifying document language for assistive technologies)
- **Charset:** Modern HTML5 standard <meta charset="UTF-8"> (universal Unicode character encoding)
- **Viewport:** <meta name="viewport" content="width=device-width, initial-scale=1.0"> (responsive viewport definition)
- **Title:** Coffi Coffee Shop Astana - Neighborhood Specialty Coffee
- **Three Key Meta Tags:**
 1. <meta name="description" content="Welcome to Coffi, a cozy neighborhood coffee shop on Alikhan Bokeikhan Street 10 in Astana, serving artisan espresso and fresh pastries.">
 2. <meta name="author" content="Samrat Zhasulan">
 3. <meta name="viewport" content="width=device-width, initial-scale=1.0">

2.2 DevTools Source Inspection (Semantic Landmark Structure)

```
DevTools Elements – Coffi Coffee Shop (index.html – Semantic HTML5 Hierarchy)

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Coffi Coffee Shop Astana - Neighborhood Specialty Coffee</title>
  <meta name="description" content="Welcome to Coffi, a cozy neighborhood coffee shop on Alikhan Bokeikhan Street 10 in Astana...">
  <meta name="author" content="Samrat Zhasulan">
</head>
<body>
  <header>
    <h1>Coffi Coffee Shop</h1>
    <nav> <!-- Semantic navigation landmark -->
      <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="about.html">About Us</a></li>
        <li><a href="menu.html">Menu</a></li>
        <li><a href="booking.html">Table Booking</a></li>
      </ul>
    </nav>
```

Element Type	Tag Name	Occurrence Count	Implementation Purpose in Coffi
Semantic Header	<header>	6	Present on all 6 pages for branding and page title
Semantic Navigation	<nav>	6	Encapsulates 6-page site navigation list on every document
Sectional Grouping	<section>	7	Thematic sectioning for visiting hours, categories, standards
Independent Article	<article>	5	Self-contained stories, bean sourcing, and code skeleton
Complementary Aside	<aside>	2	Peripheral info: visiting hours and breakfast pairing special
Data Tables	<table>	1	Relational pricing table in menu.html with caption, thead, tbody
Generic Containers	<div>	1	Strictly limited to 1 instance on menu.html:61 (legal tax note)

3. Three Structural Principles of Coffi Architecture

1. Semantic Navigation Landmarks Over Generic Containers:

- *Principle:* Instead of unsemantic wrapper divs, our Coffi site strictly encapsulates navigation links inside a semantic `<nav>` tag containing a structured `<ul>` list across all 6 pages (`index.html`, `about.html`, `menu.html`, `booking.html`, `feedback.html`, `colophon.html`).
- *Benefit:* Screen readers and search engine crawlers immediately identify navigation landmarks and jump directly between pages.

2. Explicit Label Associations and Logical Grouping in Forms:

- *Principle:* In both `booking.html` and `feedback.html`, every single input is explicitly bound to a `<label for="...">` matching the input id. Inputs are categorized into logical `<fieldset>` containers with descriptive `<legend>` headings.
- *Benefit:* Tapping or clicking any label immediately activates input focus, and assistive tools read field legends before requesting input.

3. Accessible Tabular Structure for Menu Data:

- *Principle:* In `menu.html`, beverage prices, cup sizes, and descriptions are structured in a true semantic `<table>`, featuring `<caption>`, `<thead>`, `<tbody>`, and accessible header scopes (`scope="col"`, `scope="row"`).
- *Benefit:* Screen readers announce tabular relationships, linking every drink name to its exact volume and price in tenge.

4. Hand-Drawn Page Structure Sketch

As required by Assignment 1, the website wireframe and semantic HTML5 hierarchy for Coffi Coffee Shop was planned and hand-drawn on paper with a blue ballpoint pen, detailing landmarks, section IDs, navigation flow, and signed by both students.

< header >
 Logo: Coffe (Coffee and Study Space)
 < >: [Home] [About] [Menu] [Booking] [Feedback] []

< main >
 < article >: Welcome to Coffi
 (laptop-friendly specially coffee, Bokeikhan 10)

< section id="menu" >:
 Price Table and Power-over Grid

Espresso	Flat White	Bumble
1200 ₸	1400 ₸	2200 ₸

< >:
 Hours:
 07:30-23:00
 Daily

< section id="booking" >:
 Study Desk Booking (30 seats, Wi-Fi)

< footer >
 Alikhan Bokeikhan ST 10, Astana | Tel: +7(901)880-41-04
 © 2026 Coffi Coffee Shop

Date: September 14, 2026.

Site: Coffi Coffee Shop (Astana)
 Students: Samrat Zhasulan and
 Zhaksylyk Ajilet

5. Task B — Reflective Essays

Reflective Essay: Samrat Zhasulan

A web page is fundamentally a plain-text document structured with HyperText Markup Language (HTML). At its core, an HTML file consists of human-readable elements demarcated by tags, attributes, text nodes, and entities. When a user navigates to a local HTML file or web address, the browser receives a continuous stream of raw bytes.

The rendering engine decodes these bytes into characters using the declared character encoding (UTF-8), tokenizes the stream into distinct start tags, end tags, and content, and constructs the Document Object Model (DOM) tree. Concurrently, the engine parses external stylesheets to generate the CSSOM (CSS Object Model). The browser merges DOM and CSSOM into a Render Tree, filtering out non-rendered nodes (such as `<head>` or elements with `display: none`). Next, the layout engine executes geometry calculations to determine the precise size and coordinates of each element on the viewport. Finally, the painting stage rasterizes vectors, text, borders, and colors onto physical pixels displayed on the screen. Because our assignment strictly excludes CSS and JavaScript, the browser renders directly using its default User-Agent stylesheet.

In my codebase, I strictly prioritized semantic markup over generic `<div>` containers:

1. In `menu.html`, I wrapped our pricing data inside an accessible `<table>` with `<thead>` and `<tbody>` instead of styled grid divs, giving assistive screen readers clear tabular data relationships.
2. In `menu.html`, I utilized a definition list (`<dl>`, `<dt>`, `<dd>`) instead of divs to define specialty coffee terms (Single-Origin, Bumble Coffee, Micro-foam) as term-and-description semantic pairs.
3. In `about.html`, I chose `<article>` instead of `<div>` for the community workspace narrative because it forms an independent, self-contained story that can stand on its own.

Reflective Essay: Zhaksylyk Adilet

A web page is a structured text document containing semantic tags that define document landmarks, interactive forms, and content hierarchies. When a browser opens our `booking.html` file, it parses the markup from top to bottom, building an in-memory tree representation known as the DOM (Document Object Model). Without external CSS, the browser applies its built-in user-agent style sheet to compute font sizes, element margins, and form control layouts before painting pixels to the display.

In my codebase, I strictly chose semantic elements over non-semantic `<div>` tags:

1. In `booking.html`, I grouped personal contact details and dining preferences inside `<fieldset>` elements with `<legend>` rather than wrapper divs, ensuring screen readers announce logical category contexts.
2. In `feedback.html`, I explicitly connected each `<label for="...">` with its corresponding input id instead of using generic divs with text, ensuring clicking labels activates input focus.
3. In `colophon.html`, I used `<pre>` and `<code>` instead of a div container to display sample HTML code while preserving whitespace and formatting.

When a visitor clicks the submit button on `booking.html` today, the browser triggers native client-side constraint validation against attributes like `required`, `type="email"`, `type="number"`, and `type="tel"`. If any input fails validation, the browser blocks submission and displays a native alert. Because no server-side backend script processes the request, the browser attempts an HTTP POST request to `action="#"`, reloading the static page with query fragment without transmitting data over the network.

6. AI Assistance & Usage Logs

Interactive logs documenting prompt history, academic integrity declarations, and AI-assisted workflow iterations are maintained in the GitHub repository:

- Samrat Zhasulan — AI Usage Log: <https://github.com/jasulansam/web/blob/main/ai-log.md>
- Zhaksylyk Adilet — AI Usage Log: <https://github.com/jasulansam/web/blob/main/ai-log.md>

7. W3C Validation Evidence

All six project documents were submitted to the official W3C Nu HTML Validator, passing with zero errors and zero warnings.

1. val-index (index.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Welcome to Coffi, a cozy neighborhood coffee shop on Alikhan Bokeikhan Street 10 in Astana, serving ar">
  <meta name="author" content="Zamrat Zhasulan">
  <!-- Author: Zamrat Zhasulan -->
  <title>Coffi Coffee Shop Astana - Neighborhood Specialty Coffee</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

[About this checker](#) • [Report an issue](#)

2. val-about (about.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Discover the story, bean sourcing, and study-friendly atmosphere of Coffi coffee shop on Alikhan Bokeil">
  <meta name="author" content="Zamrat Zhasulan">
  <!-- Author: Zamrat Zhasulan -->
  <title>About Coffi - Coffee Philosophy and Atmosphere</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

[About this checker](#) • [Report an issue](#)

3. val-menu (menu.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Explore the full menu and pricing at Coffi Astana, featuring specialty espresso, cold bumble drinks, m
  <meta name="author" content="Zamrat Zhasulan">
  <!-- Author: Zamrat Zhasulan -->
  <title>Menu & Price List - Coffi Coffee Shop Astana</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.

Total execution time 4 milliseconds.

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4. val-booking (booking.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Reserve a table or dedicated quiet study desk at Coffi on Alikhan Bokeikhan Street 10 in Astana.">
  <meta name="author" content="Zhaksylyk Adilet">
  <!-- Author: Zhaksylyk Adilet -->
  <title>Table & Workspace Reservation - Coffi Coffee Shop</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.

Total execution time 4 milliseconds.

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5. val-feedback (feedback.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Share your thoughts and feedback about your coffee and workspace experience at Coffi on Alikhan Bokeik">
  <meta name="author" content="Zhaksylyk Adilet">
  <!-- Author: Zhaksylyk Adilet -->
  <title>Customer Feedback & Reviews - Coffi Coffee Shop</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

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6. val-colophon (colophon.html — Passed, 0 Errors)

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Colophon documenting technical architecture, markup rules, and validation details for Coffi website.">
  <meta name="author" content="Zhaksylyk Adilet">
  <!-- Author: Zhaksylyk Adilet -->
  <title>Colophon & Technical Architecture - Coffi Website</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 4 milliseconds.

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8. HTML5 Semantic Elements Checklist

The comprehensive line-by-line checklist of every required HTML5 semantic tag, attribute, list, and table structure with student attributions is documented in the repository:

URL: <https://github.com/jasulansam/web/blob/main/tag-checklist.md>